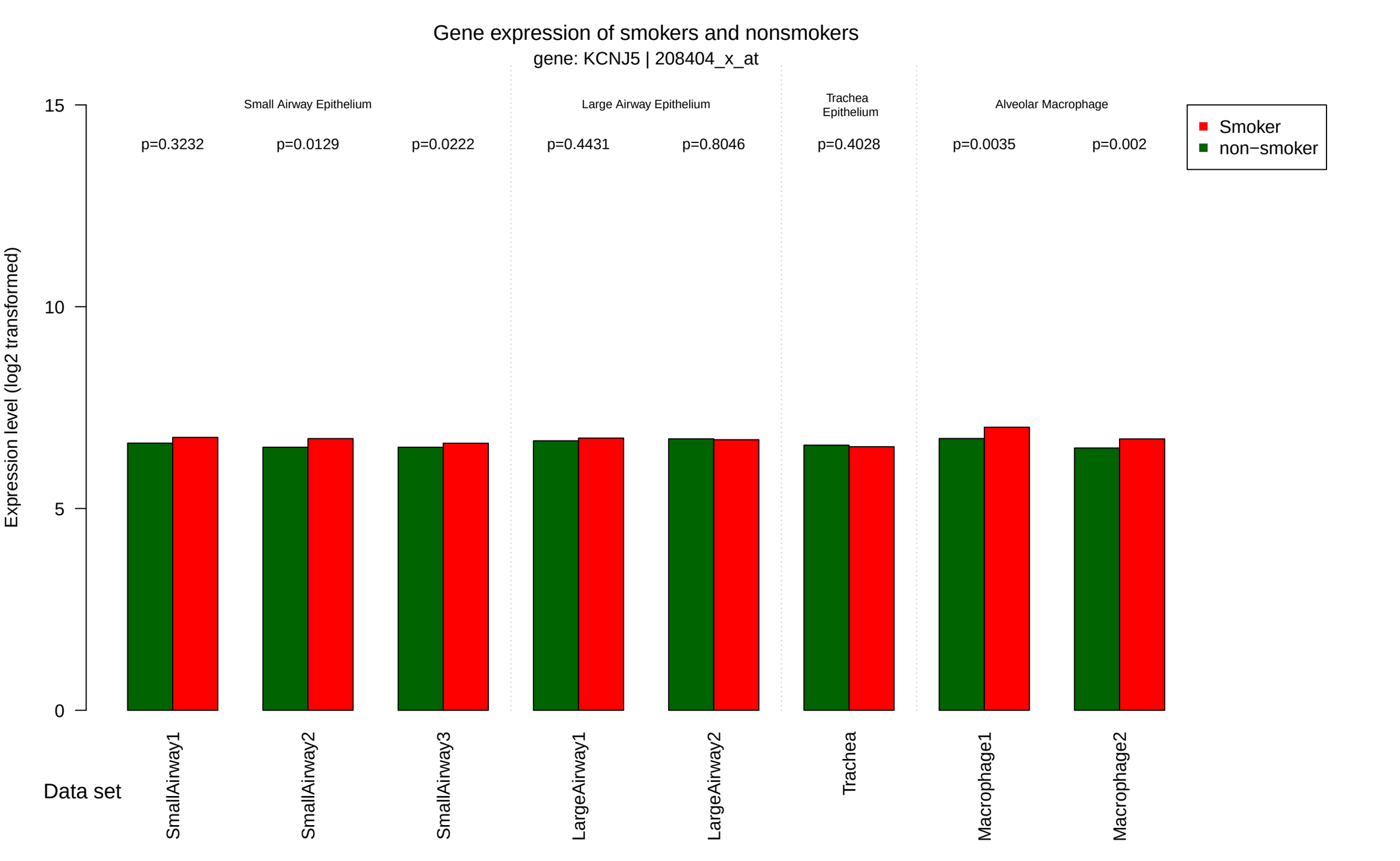
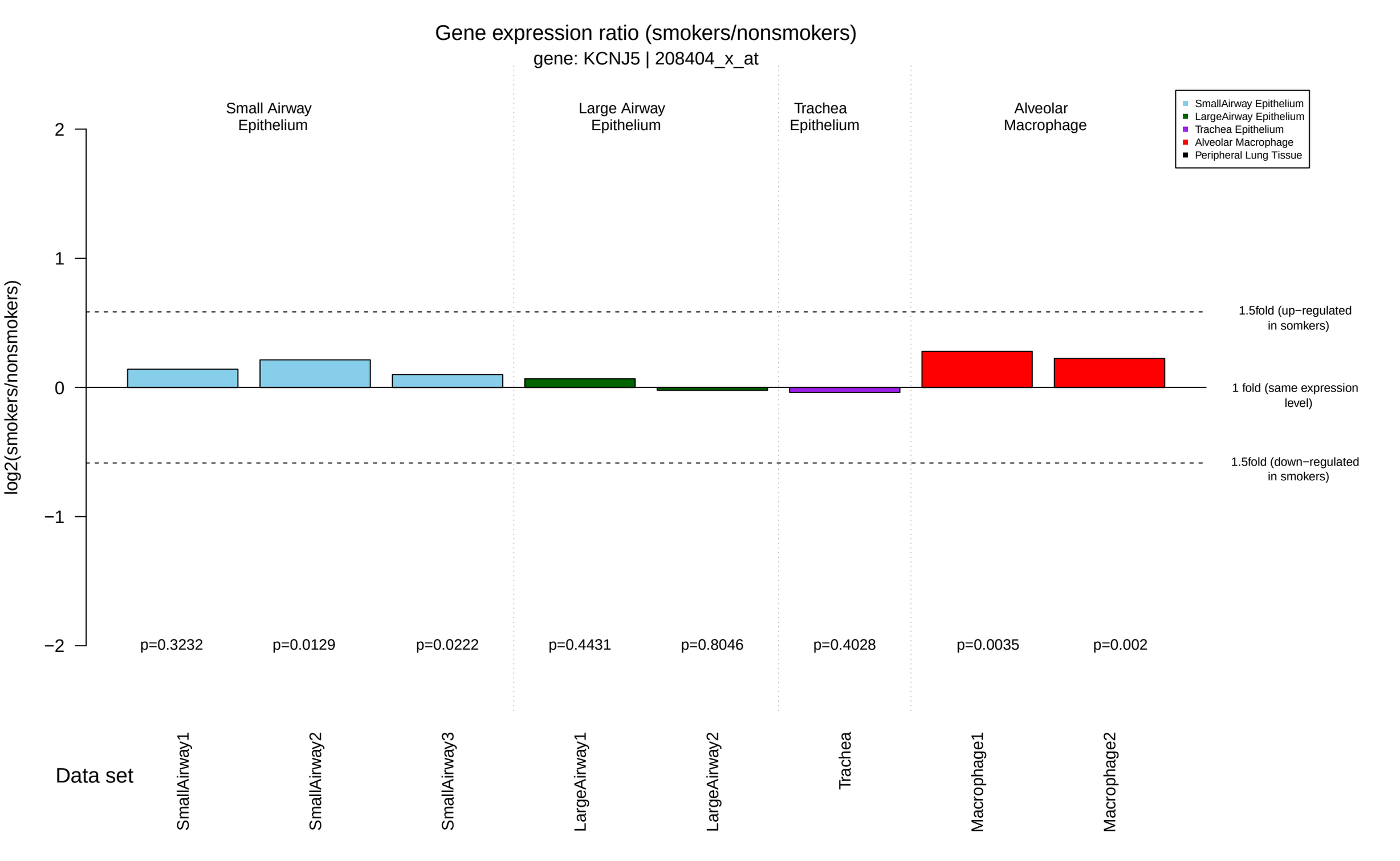
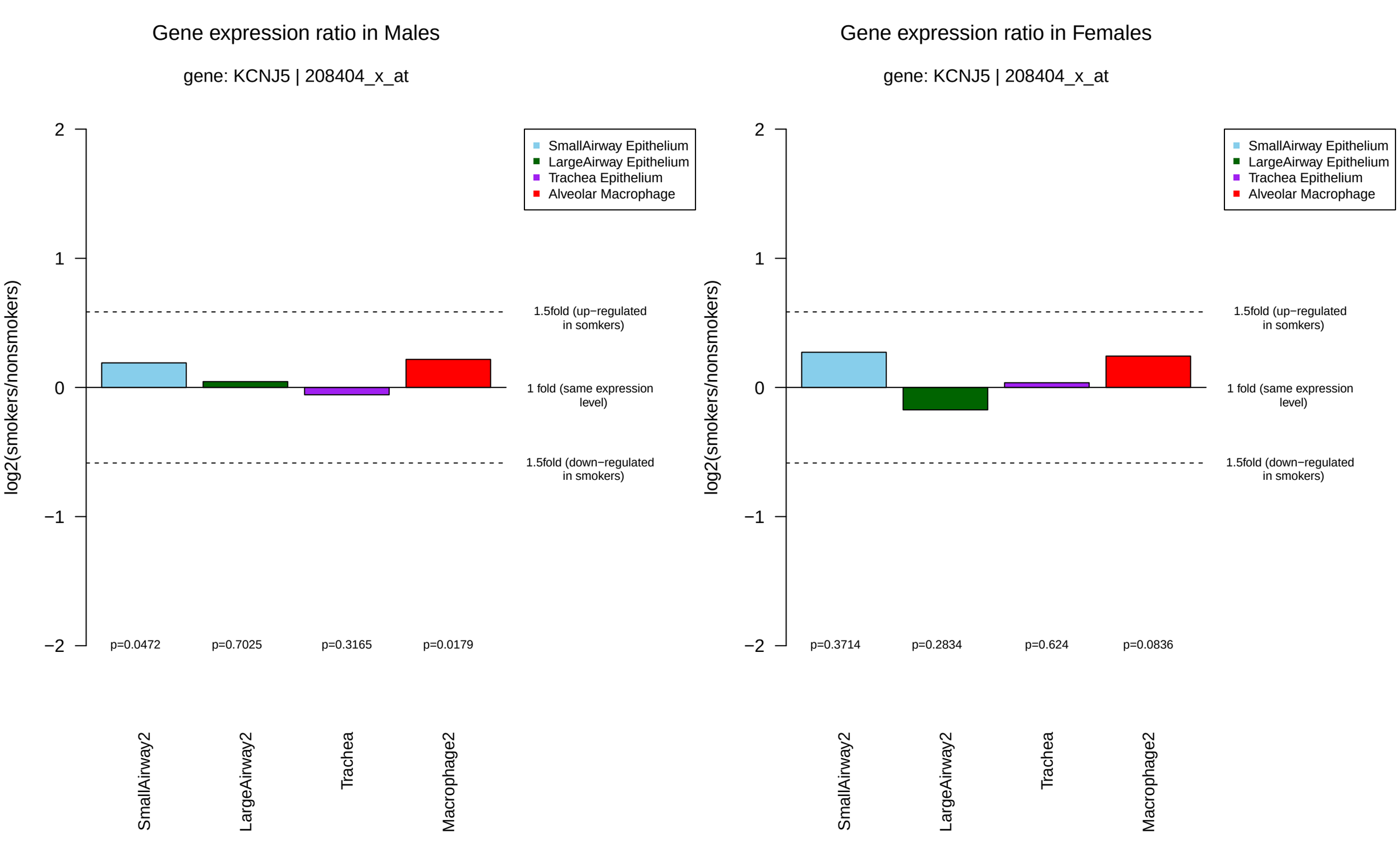
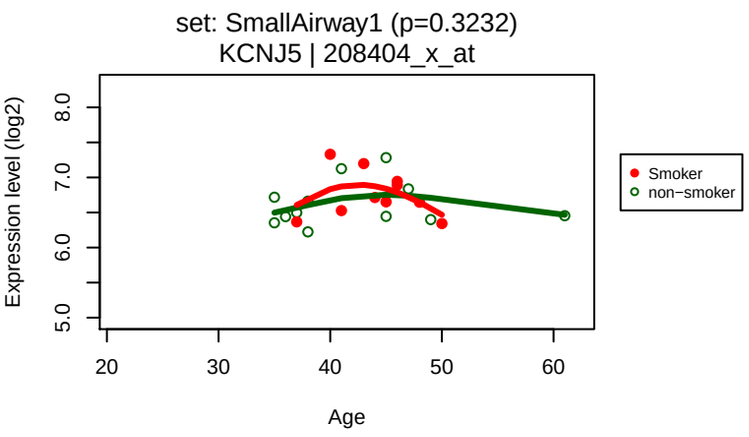
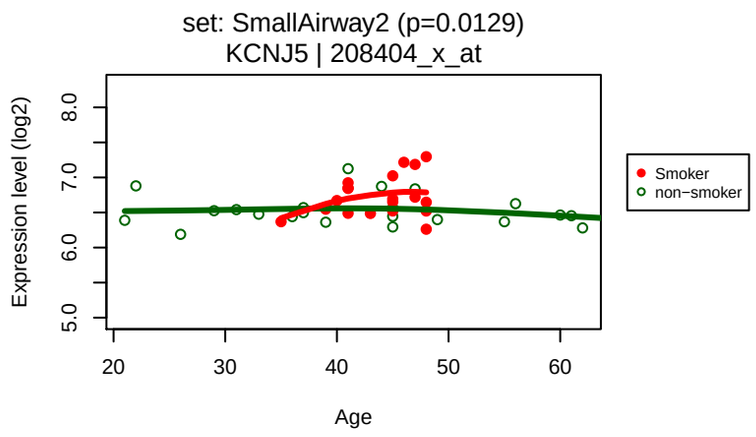
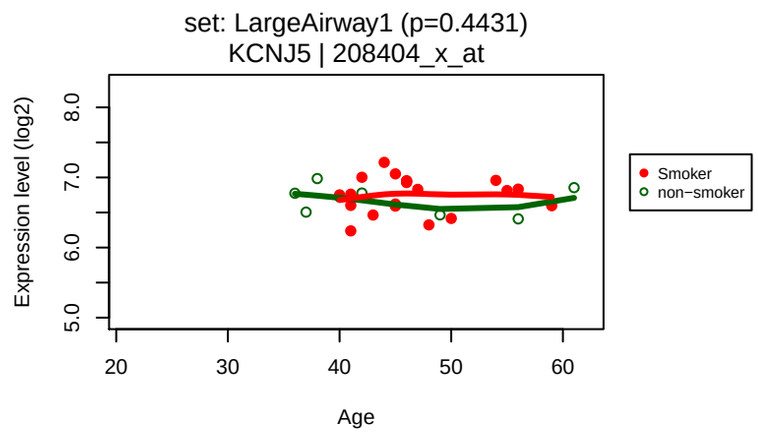

Gene expression of smokers and nonsmokers
gene: KCNJ5 | 208404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3232
p=0.0129
p=0.0222
p=0.4431
p=0.8046
p=0.4028
p=0.0035
p=0.002
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

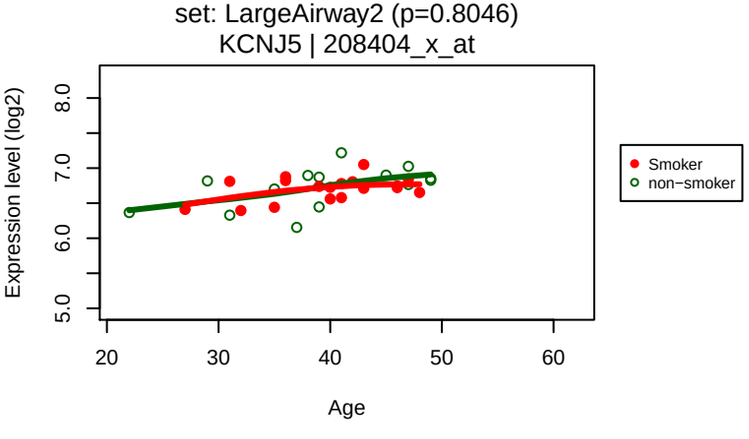

Gene expression ratio (smokers/nonsmokers)
gene: KCNJ5 | 208404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3232
p=0.0129
p=0.0222
p=0.4431
p=0.8046
p=0.4028
p=0.0035
p=0.002
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

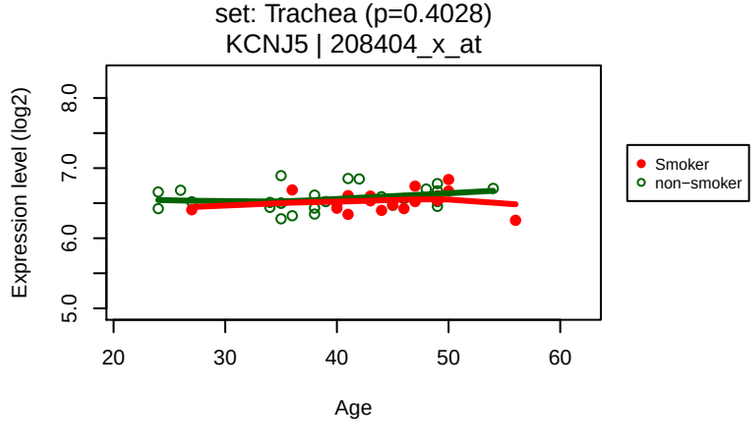

Gene expression ratio in Males
gene: KCNJ5 | 208404_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0472
p=0.7025
p=0.3165
p=0.0179
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNJ5 | 208404_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3714
p=0.2834
p=0.624
p=0.0836
SmallAirway2
LargeAirway2
Trachea
Macrophage2

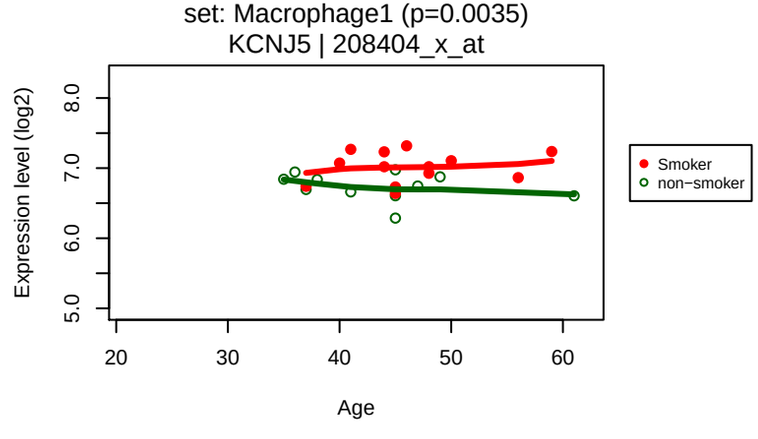

set: SmallAirway1 (p=0.3232)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

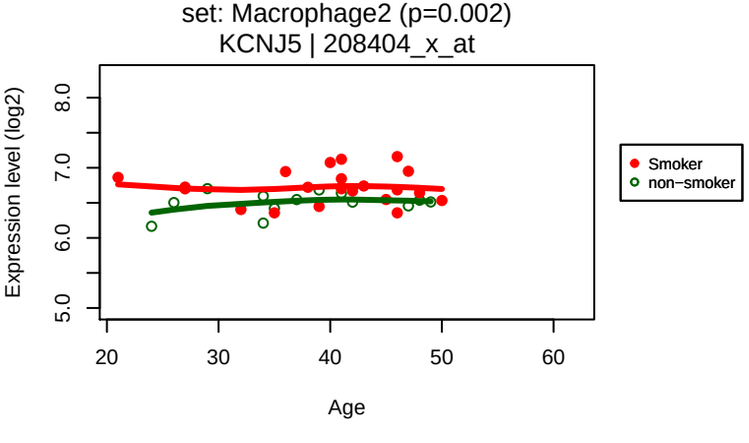

set: SmallAirway2 (p=0.0129)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

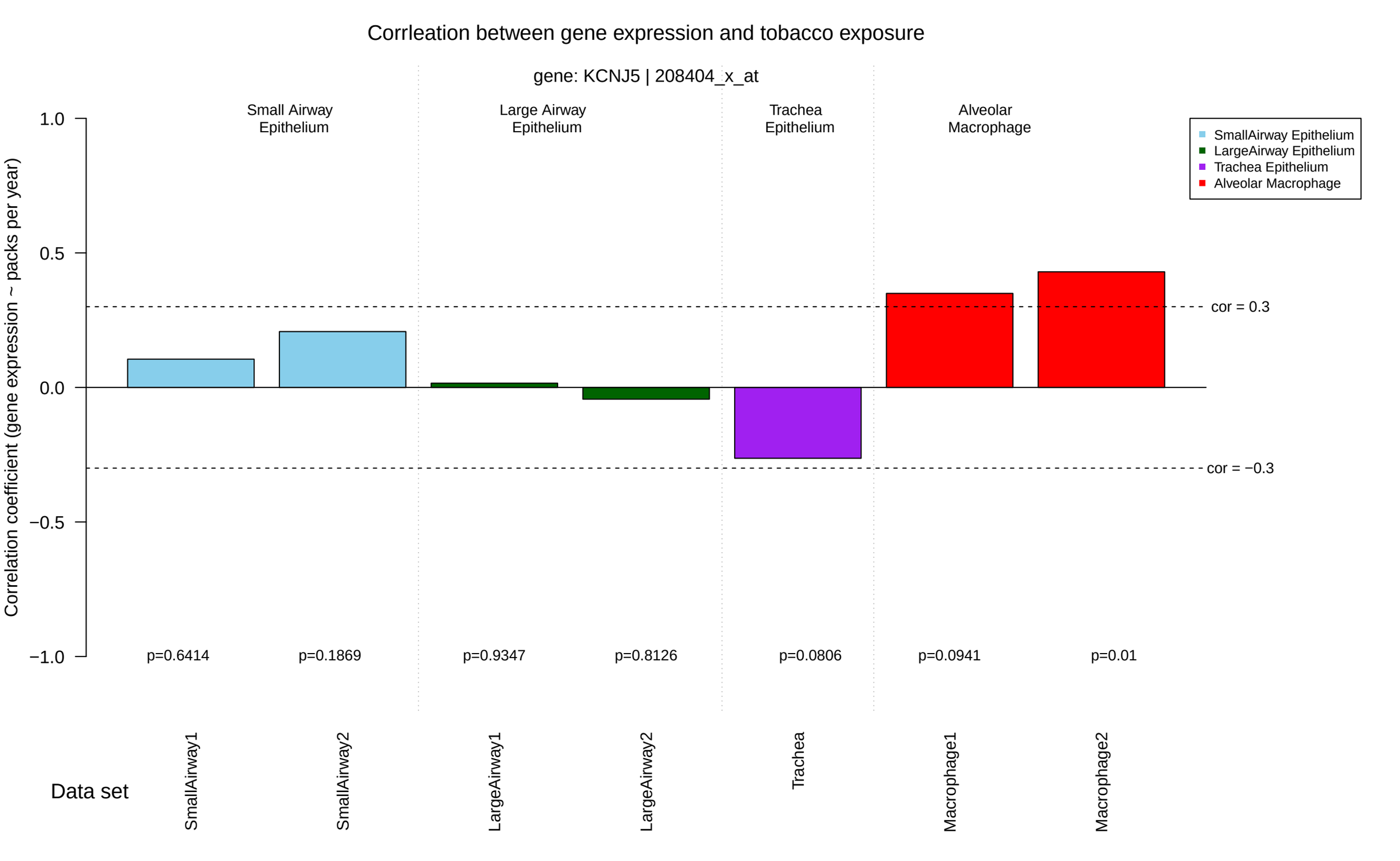

set: LargeAirway1 (p=0.4431)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8046)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4028)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0035)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.002)
KCNJ5 | 208404_x_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNJ5 | 208404_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6414
p=0.1869
p=0.9347
p=0.8126
p=0.0806
p=0.0941
p=0.01
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2